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Glazing with lighting capabilities

Abstract

This disclosure relates to a glazing with a laminated structure for providing lighting capabilities that comprises a first glass sheet with a first surface and a second surface, a second glass sheet with a third surface and a fourth surface, an interlayer between the first glass sheet and the second glass sheet, and a lighting system, which may include a light source, a light guide attached to the first light source and at least partially laminated between the first glass sheet and the second glass sheet, and a light extraction zone having a light extraction means laminated between the first glass sheet and the second glass sheet.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a National Stage of International Application No. PCT/US22/81412 filed on Dec. 13, 2022 which claims priority to and benefit of U.S. Provisional Patent App. No. 63/289,374 filed Dec. 14, 2021, titled GLAZING WITH LIGHTING CAPABILITIES, U.S. Provisional Patent App. No. 63/289,382 filed Dec. 14, 2021, titled GLAZING WITH LIGHTING CAPABILITIES, U.S. Provisional Patent App. No. 63/289,400 filed Dec. 14, 2021, titled GLAZING WITH LIGHTING CAPABILITIES, and U.S. Provisional Patent App. No. 63/289,405 filed Dec. 14, 2021, titled GLAZING WITH LIGHTING CAPABILITIES, the entire contents of each of which are incorporated by reference herein in their entirety and relied upon.

BACKGROUND

(1) The present disclosure is generally related to a glazing with lighting capabilities. There are numerous reasons an automotive vehicle may need to supply light to the interior or the exterior of the vehicle. For example, a rear windshield may be equipped with a rear brake light that emits light to the exterior of the vehicle whenever the brakes of the vehicle are engaged. Alternatively, lighting may be provided to the interior cabin of a vehicle to supply reading light or ambient lighting. Current vehicle cabin lighting systems may focus on providing large, centrally located cabin lights or overhead passenger lights that do not supply the desired light that some passengers require. For example, cabin lights may provide a light that is too intense for some passengers. Additionally, physical design constraints in automotive vehicles mean having the flexibility to supply the required light to the interior or exterior through different space efficient and aesthetically pleasing means is attractive.

(2) For each of the above reasons, it is desirable to provide an automotive glazing with lighting capabilities for providing ambient lighting for a vehicle.

SUMMARY

(3) The laminated glazing disclosed herein improves on existing glazing by providing a glazing with an integrated lighting system for providing light of a desired intensity to a desired location along the glazing and throughout the cabin of the vehicle.

(4) In light of the disclosure, and without limiting the scope of the invention in any way, in a first aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a glazing comprises a laminated structure including a first glass sheet with a first surface and second surface, a second glass sheet with a third surface and fourth surface, an interlayer between the first glass sheet and the second glass sheet, and a lighting system. The lighting system includes at least one first light source and at least one first light guide attached to each first light source. The lighting system also includes at least one first light extraction zone for directing light into a second light guide within the laminated structure. The second light guide may include a second extraction zone, including a second light extraction means wherein light in the second light guide is extracted by the second light extraction means at the second light extraction zone.

(5) In a second aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the glazing is a sunroof.

(6) In a third aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the second light extraction means is on a surface of the first glass sheet or the second glass sheet.

(7) In a fourth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the second light extraction means is on a film.

(8) In a fifth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the film is laminated between a first glass sheet and a second glass sheet.

(9) In a sixth aspect of the present disclosure, which may be combined with any other aspect listed

herein unless specified otherwise, the film is applied to a surface of the first glass sheet or the second glass sheet.
(10) In a seventh aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the second extraction means includes a print
(11) In an eighth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the second extraction means is printed on a surface of the first glass sheet or the second glass sheet.
(12) In a ninth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the second extraction means is printed on a surface of the interlayer.
(13) In a tenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the second light guide is the second glass sheet and the second glass sheet faces a vehicle interior when installed.
(14) In an eleventh aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first light extraction means includes a luminescent composition.
(15) In a twelfth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first light source is a laser.
(16) In a thirteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first light guide at least partially or completely surrounds the laminated glazing.
(17) In a fourteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first light guide and the first light extraction zone are adhered to an edge of the laminated structure and an encapsulation material is applied around the laminated structure edge, over the first light guide and the first light extraction zone.
(18) In a fifteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first light guide may be at least partially laminated between the first glass sheet and the second glass sheet and the first light extraction zone may be laminated between the first glass sheet and the second glass sheet.
(19) In a sixteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the light absorption coefficient in the second light guide is less than 0.1/cm.
(20) In a seventeenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first light extraction zone has at least one area of a variable light intensity.
(21) In a eighteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, there are three first light guides and three corresponding light sources and first light extraction zones.
(22) In a nineteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the three first light extraction zones are three separate colors: red, green, and blue.
(23) In a twentieth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, multiple first light extraction zones each include a different light extraction means.
(24) In a twenty-first aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the laminated structure includes a switchable layer.
(25) In a twenty-second aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the laminated structure of the glazing further includes an infrared reflective layer.
(26) In a twenty-third aspect of the present disclosure, which may be combined with any other

aspect listed herein unless specified otherwise, the infrared reflective layer is a coating provided on the first glass sheet.
(27) In a twenty-fourth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the infrared reflective layer is provided as a film laminated between the first glass sheet and the switchable layer.
(28) In a twenty-fifth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the laminated structure of the glazing further includes a low-emissivity coating on the second glass sheet.
(29) In a twenty-sixth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the first interlayer is darkened in color.
(30) In a twenty-seventh aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a glazing comprises a first glass sheet with a first surface and second surface and a lighting system. The lighting system includes at least one first light source and at least one first light guide attached to each first light source and at least partially or completely surrounding the first glass sheet. The lighting system also includes at least one first light extraction zone for directing light into a second light guide within the laminated structure. The second light guide may include a second extraction zone, including a second light extraction means wherein light in the second light guide is extracted by the second light extraction means at the second light extraction zone.
(31) Additional features and advantages are described in, and will be apparent from, the following Detailed Description and the Figures. The features and advantages described herein are not all-inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the figures and description. Also, any particular embodiment does not have to have all of the advantages listed herein and it is expressly contemplated to claim individual advantageous embodiments separately. Moreover, it should be noted that the language used in the specification has been selected principally for readability and instructional purposes, and not to limit the scope of the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The accompanying drawings, which are incorporated into and constitute a part of this specification, illustrate one or more example aspects of the present disclosure and, together with the detailed description, serve to explain their principles and implementations.
(2) FIG. **1** illustrates a cross section of a glazing according to an exemplary embodiment of the present disclosure.
(3) FIG. **2** illustrates a lighting system of a laminated glazing according to an exemplary embodiment of the present disclosure.
(4) FIG. **3** illustrates a lighting system of a laminated glazing according to an exemplary embodiment of the present disclosure.
(5) FIG. **4** illustrates a cross section of a laminated glazing according to an exemplary embodiment of the present disclosure that includes a first light guide that is at least partially laminated between two layers of an interlayer.
(6) FIG. **5** illustrates a cross section of a laminated glazing according to an exemplary embodiment of the present disclosure that includes multiple first light guides and multiple first light extraction zones.
(7) FIG. **6** illustrates a cross section of a laminated glazing according to an exemplary embodiment of the present disclosure.
(8) FIG. **7** illustrates a cross section of a laminated glazing according to an exemplary embodiment

of the present disclosure including an infrared reflective layer and low emissivity coating.
(9) FIG. **8** illustrates a cross section of a laminated glazing according to an exemplary embodiment of the present disclosure that includes a single glass sheet.

DETAILED DESCRIPTION

(10) In the following description, for purposes of explanation, specific details are set forth in order to promote a thorough understanding of one or more aspects of the disclosure. It may be evident in some or all instances, however, that any aspects described below can be practiced without adopting the specific design details described below.
(11) Among other features, the present disclosure provides an automotive glazing having lighting capabilities. Particularly, the automotive glazing may provide ambient lighting to a vehicle interior. Such ambient lighting may preferably be found above a passenger compartment of the vehicle, such as in a sunroof. The ambient lighting may provide an aesthetic function within the vehicle. In some embodiments, the laminated glazing described herein may be used in other automotive glazings, such as side windows or partitions.
(12) FIG. **1** illustrates a cross section of a glazing according to an exemplary embodiment of the present disclosure. The glazing **100** has a laminated structure **110**, including a first glass sheet **101** with a first surface **111** and second surface **112**, a second glass sheet **102** with a third surface **113** and fourth surface **114**. For example, the glazing **100** may be designed for use as a sunroof of a vehicle, where the first surface **111** of the first glass sheet **101** faces, such that it is exposed to, the exterior of the vehicle, the fourth surface **114** of the second glass sheet **102** faces the interior of the vehicle, and both the second surface **112** and third surface **113** face an interlayer **103**, **104**. The first and second glass sheets **101**, **102** may preferably be from 0.5 mm to 3 mm, more preferably from 1 mm to 2.5 mm. The glass sheets **101**, **102** may include, without limitation, soda-lime silicate glass described by ISO 16293-1:2008. The laminated structure **110** of the glazing **100** may include an interlayer, which may contain one or more layers, such as a first interlayer **103** and a second interlayer **104**. The interlayer is laminated between the second surface **112** of the first glass sheet **101** and the third surface **113** of the second glass sheet **102**. The interlayer **103**, **104** may include an adhesive polymer film, such as a polyvinyl butyral, or an ionomer. The laminated structure **110** may have at least four sides. The glazing may also include a lighting system **120** that features a first light guide **122** and a second light guide **132**, for example for providing a cabin light in a vehicle.
(13) FIG. **2** illustrates an example of a lighting system of an embodiment of the present disclosure. A laminated glazing **100** according to an exemplary embodiment of the present disclosure may include a lighting system **120**. A lighting system **120** for providing a light via a laminated glazing **100** may include a first light source **121** attached to a first light guide **122** and containing a first light extraction zone **123**. The first light source **121** may be detachable from the first light guide **122**. The first light source **121** may couple to the first light guide **122**, for example, by a fiber optic connector or optical fiber coupler. The first light source **121** may preferably include a laser light source.
(14) The light from the light source **121** may be directed to a first light guide **122**. The light may reflect within the first light guide **122** with total internal reflection before reaching a light extraction zone **123**. The first light guide **122** may have an index of refraction higher than that of the surrounding materials. The first light guide **122** may be thin and may be flexible so as to wrap around an edge of the laminated structure **110**, as shown in FIG. **2**. The light guide **122** may, for example, include a glass or polymer based optical fiber having a thickness of 0.1 to 1.5 mm.
(15) The first light guide **122** may have a first light extraction zone **123** where light is emitted from the first light guide **122** into the laminated structure **110** of the glazing **100**. In some embodiments, the first light extraction zone **123** may preferably surround or partially surround the laminated structure **110** so as to introduce light around the edge of the laminated structure **110** of the glazing **100**, as shown in FIG. **2**. For example, the first extraction zone **123** could substantially surround the laminated structure **110** of a windshield in order to provide light along the edges of the laminated

structure **110** in order to provide an aesthetically pleasing ambient light to the cabin of a vehicle. In some embodiments, the first light extraction zone **123** may extend over only part of the glazing **100** edge. For example, the first light extraction zone **123** may extend along fewer than all of the glass edges using one or more first light guides **122**. The light extraction zone **123** may also be provided in multiple separated portions of the light guide **122** such that there may be portions of the light guide **122** extending between sections including the light extraction zone **123**. The first light guide **122** and the first light extraction zone **123** may be positioned around the edge of the laminated structure **110** and adhered to the structure **110**. A resin adhesive may be preferable in some embodiments.

(16) The light emitted from the first light extraction zone **123** may be extracted from a portion of the first light guide **122**. Particularly, where the first light guide **122** is a round fiber, the first light extraction means may not extend around the entire fiber. Further, if the first light guide is positioned in a design which includes twists or turns, the positioning of the first light extraction zone **123** may be determined to compensate for the shape of the first light guide **122**. The light guide **122** may be provided around or near an edge of laminated glass sheets **101**,**102** or may be provided in any desired area of the glazing, such as in a design towards a middle of the glazing.

(17) The first light extraction zone **123** may include a first light extraction means, which may include an area of a fiber first light guide **122** having an extraction composition, such as a luminescent composition which may include luminescent particles or other luminescent material, thereon. A luminescent composition may include appropriate luminescent materials to extract the light in a desired visible wavelength. For example, the light may be extracted as white colored visible light. The intensity of the light may depend on the power supplied to the light source **121** in the lighting system **120**. The power supplied to the light source **121** may be controlled by an electrically controlled driver unit. A user may be able to adjust the intensity of the light with a dimming switch.

(18) The light may be extracted at the first light extraction zone **123** into a second light guide **132**. The second light guide **132** may include a glass sheet or another layer of the laminated structure **110**. For example, in some embodiments, the second light guide **132** may be a film laminated within the laminated structure (for example, as shown in FIG. **1**). A second light guide **132** implemented as a layer of the laminated structure **110** preferably is clear or substantially clear in color to prevent loss of light by absorption through the second light guide **132** which would otherwise occur during reflection of the light through the second light guide **132**.

(19) On a pathlength x through the light guide, the light intensity may decay according to I(x)=I.sub.0 exp(−αx) with light absorption coefficient α. Preferably the light absorption coefficient is less than 0.1/cm through the second light guide **132**, more preferably less than 0.05/cm. The light may enter the second light guide **132** at an angle suitable for total internal reflection so that the light may reflect through the second light guide **132** to a second light extraction zone **133** having a second light extraction means. The second light guide **132** may be surrounded by materials having an index of refraction lower than the second light guide **132**.

(20) The second light guide **132** may include a second light extraction zone **133** which may include a second light extraction means for providing an ambient light to a vehicle interior. The second light extraction zone **133** may be designed and positioned to complement a design of the first light guide **122** and first light extraction zone **123** in some embodiments. The second light extraction means may be diffusive, reflective, or combinations thereof. A diffusive second light extraction means may be preferable. Where a pattern in the ambient light provided is desired, the second light extraction means may be varied across the second light extraction zone **133** to provide such a pattern. For example, a concentration of diffusive and/or reflective materials may be varied across the second light extraction zone **133**. It may be preferable to provide a symmetrical dispersion of light across the glazing **100**. A variation in the second light extraction zone **133** may also be used to provide an even appearance across a glazing **100**. Light may dull moving away from the first light

extraction zone **123**. If an even light dispersion is desired across a glazing **100**, the second light extraction means may be varied to provide less diffusion near the first light extraction zone **133** and more diffusion away from the first light extraction zone **133**. Such variation may be achieved, for example, by a change in the density of diffusive particles across the second light extraction zone **133** or by using a combination of different materials.

(21) A second light extraction means may include a coating or printed material, such as a non-opaque enamel or a fine particle containing coating, or a diffusive material embedded in the second light guide **132**. The fine particles may have an index of refraction higher or lower than the second light guide **132**. The second light extraction means may, in some further embodiments, include roughening of a glass surface, which may include treating the glass surface with a laser. Light extracted by the second light extraction means may preferably be a diffused light for providing a desired ambient light to a vehicle interior. For example, the second light extraction means may be positioned on the glazing **100**, including on a layer of the laminated structure **110** glazing **100** so as to provide diffused light to a vehicle interior. The second light extraction means may be provided on the surface of the second light guide **132**, such as a glass surface, an interlayer surface, or a film surface, and may be positioned within the laminated structure **110** or on an outer surface of the laminated structure **110**. The second light extraction means may be provided as a coating, a paint, or a film, for example, and may be provided over a part of the glazing, as shown in FIG. **2**, or an entire glazing surface. For example, where the second light extraction means is provided over part of the glazing **100**, the second light extraction means may provide a pattern across the glazing. Additionally, a second light extraction means may be positioned on a film, which may include a laminated film or a film, on the third surface **113** of the second glass surface **102**, for example, facing a vehicle interior when the glazing is installed.

(22) FIG. **3** illustrates another example of a lighting system of an embodiment of the present disclosure. A lighting system **120** of an embodiment of the present disclosure may include a second light guide **132** implemented as a glass sheet or film laminated between the first and second glass sheets **101**, **102**. In some further embodiments, the second light guide **132** may be a layer of the laminated glazing **100**, such as the second glass sheet **102**. The second light extraction zone **133** which includes a second light extraction means may be attached to the second light guide **132**. The second light extraction means may be provided as a coating or a film provided over the entire glazing surface, as shown in FIG. **3**.

(23) The positioning of the first light extraction zone **123** may cause local increases or decreases in light intensity where the first light extraction zone **123** turns at a corner or in a design. This may occur where portions of the first light extraction zone **123** are close to each other and relatively more light is being distributed than in areas without such proximity or where a positive corner causes a relative decrease in light intensity. Intensity correction may be achieved with adjustments to the first light extraction means. The first light extraction means may affect the local intensity of the light provided via the lighting system. The first light extraction zone **123** and first light extraction means may, for example, be adjusted at a corner of the laminated structure for reducing light intensity towards a corner. The change in intensity may preferably be gradual towards and away from a corner of the laminated structure with a corner having relatively lowest light intensity. Without such changes to the light, there may be an uneven brightness of the light to an observer as the sides of light would come together closely in the corners. Thus, the fading light towards a corner may compensate for this so that the light from each side of a corner may combine when close to the corner and an observer may see a more even light distribution around the glazing. The light output from the first light extraction zone **123** may be locally increased where there is a positive corner which otherwise would provide a space of relatively dimmer light. The first light extraction means (and light intensity) may be adjusted, for example, by altering the local concentration of luminescent material in the first light extraction means. The light intensity may also be altered as necessary for evening of light distribution where the first light extraction zone is

provided in a decorative pattern or in other areas that would otherwise provide an uneven light output. In some embodiments, it may be desired to create a lighting design of changing light distribution with such a variation in light output from the first light extraction zone **123**.
(24) FIG. **4** illustrates a cross section of an example embodiment of a glazing of the present disclosure. A lighting system **120** for providing lighting capabilities to a laminated glazing may be substantially laminated within the laminated structure **110**. For example, at least a portion of the first light guide **122** and the first light extraction zone **123** may be laminated between the first glass sheet **101** and the second glass sheet **102**. In some embodiments, the laminated first light guide **122** and first extraction zone **123** may be laminated between two interlayer layers **103**, **104**. Where at least some of the lighting system **120** is laminated between the first and second glass sheets **101**, **102**, the first light source **121** may be detachable such that it can be attached to the first light guide **122** after lamination. The first light guide **122** may extend out of the interlayers **103**, **104** and the glazing **100** for attachment to a first light source **121**. In some embodiments, the first or second glass sheet **101**,**102** may include a recess or an opening through which an optical fiber first light guide **122** may pass. The first light guide **122** and first light extraction zone **123** may be in any desired shape. The first light guide **122** may be a material, including an optical fiber, which may be flexible to follow a shape of the glazing **100** or provide another design in the glazing **100**.
(25) The first light extraction zone **123** may be positioned along an area of the laminated structure having an opaque masking layer **105**. The opaque masking layer **105** may include an opaque print around the periphery of the laminated structure **110**. Such an opaque print may be an enamel print, which may be black, and may be provided on a surface of the first glass sheet **101** and/or the second glass sheet **102**. The opaque masking layer **105** may block a view of the first light extraction zone **123**. In some embodiments, the opaque masking layer **105** may include additional blocking materials, such as a metallic layer, to prevent or limit visibility of the first light extraction zone. The second light extraction zone **133** may include the second light extraction means.
(26) FIG. **5** illustrates a cross section of an example embodiment of a glazing of the present disclosure. A lighting system **120** for providing lighting capabilities to a glazing **100** having a laminated structure **110** may include multiple light sources, first light guides **122**, and first light extraction zones **123**. For example, the multiple first light sources **121***a-c* may be provide different colors of light. The multiple first light extraction zones **123***a-c* may have different compositions to provide multiple color options to a user. Particularly, the first light extraction zones **123***a-c* may emit red, green, and blue light, respectively. The first light sources **121***a-c* may be connected to a control mechanism for controlling the color output, including relative light intensity, to provide a desired appearance. The red, green, and blue light may be directed from the first light extraction zones **123***a-c* to the second light guide **132** (second glass sheet **102** in FIG. **5**) where the light may be internally reflected until reaching a second light extraction zone **133** which includes the second light extraction means. The light may be extracted to a vehicle interior at the second light extraction zone **133**. The multiple first light guides **122***a-c* and first light extraction zones **123***a-c* may be aligned with each other and positioned together within the construction. It may be preferable to provide the multiple first light guides **122***a-c* in a stacked formation to allow the light extraction from each of the first light extraction zones **123***a-c*. In some embodiments, the first light guides **122** may be twisted together and the corresponding first light extraction zones may be positioned to account for the twisting. The multiple light sources may be contained in separate housings or may be all positioned within one housing. Where the first light guides **122***a-c* and first light extraction zones **123***a-c* are positioned around the edge of a laminated structure **110**, the second glass sheet **102** may have a thickness suitable for placement of the three first light guides and first extraction zones, as shown in FIG. **5**. The laminated structure **110** of the glazing **100** having such a lighting system **120** may preferably include clear or light colored layers between a vehicle interior and the second light extraction zone **133** to provide a suitable appearance of the light emitted for ambient lighting. Some embodiments may include the use of tinted glass,

interlayers, coatings, or films outside of the second light extraction zone **133**. Some embodiments may further include a solar control layer within the interlayer and outside of the second light guide **132** and second light extraction zone **133**. A solar control layer may include, for example, an infrared reflective layer or layers, which may be provided as a coating on a glass surface or film.
(27) FIG. **6** illustrates a cross section of another example embodiment of a laminated glazing of the present disclosure. The laminated structure **110** of a glazing **100** of the present disclosure may further include a switchable layer **107**. A switchable layer **107** may include an electrically controllable layered material which may change in opacity based on an electrical signal. The switchable material may include a film laminated between first and second glass sheets **101**, **102**. Switchable films may include a liquid crystal film, which may include substrate layers each having an electrode thereon sandwiching a liquid crystal core layer. The switchable film may be laminated between interlayers **103**, **104** between first and second glass sheets **101**, **102**. The interlayers **103**, **104** may be adhesive layers and may be the same or different in thickness and composition. The first light extraction zone **123** may emit light to a second light guide **132** that is between the switchable layer and the vehicle interior when installed in a vehicle, particularly between the switchable layer and an outer surface (e.g. the fourth surface **114** of the second glass sheet **102**), as shown in FIG. **6**.
(28) In other example embodiments, the second glass sheet **102** and/or the second interlayer **104** may serve as the second light guide **132**. In some embodiments, a second light guide **132** may include a film laminated between the switchable layer **107** and the second glass sheet **102**. The second glass sheet **102** may preferably be clear in color so as to not limit the light emitted by the lighting system. The second light extraction zone **133** includes a second light extraction means, which may be provided on a surface of the second glass sheet **102**, as shown in FIG. **6**, a surface of the second interlayer, or a surface of the switchable layer. In some embodiments, the second light extraction means may include diffusing particles embedded within a layer in the glazing. As in other example embodiments, the second light extraction means may cover all or part of the glazing **100**. In some embodiments, it may be preferable to include a darkened layer to the exterior of the switchable layer **107**. For example, the first glass sheet **101** or the first interlayer **103** may be darkened in color, which may include a green or grey color.
(29) FIG. **7** illustrates a cross section of another example embodiment of a laminated glazing of the present disclosure that includes a switchable layer **107**. A laminated glazing **100** having a switchable layer **107** may include a darkened layer, which may include a glass sheet **101** or an interlayer **103**,**104** on either side of the switchable layer **107**. Where an interlayer **104** between a switchable layer **107** and a vehicle interior is darkened, an additional interlayer may be provided between the darkened interlayer **104** and the second glass sheet **102** to prevent loss of light when reflected within the second light guide **132** (which may be the second glass sheet **102**). The laminated structure **110** of a glazing **100** may include a solar control layer. A solar control layer may include, for example, an infrared reflective layer **108** or layers, which may be provided as a coating on a glass surface or film. Where an infrared reflective layer **108** is included, the first glass sheet may preferably be clear in color. The infrared reflective layer **108** may be included to improve the overall thermal comfort of a vehicle by limiting the infrared light reaching a vehicle interior. Such a layer is preferably provided to an exterior of the switchable layer **107** where there is a switchable layer **107** and may also protect the switchable layer **107** from infrared light. The infrared reflective coating **108** may preferably be provided on a surface of the first glass sheet **101** and an infrared reflective film may be laminated between interlayers **103**, **104**, between the first glass sheet **101** and the switchable layer **107**. To provide a dark color to the laminated glazing **100** having an infrared reflective layer **108**, an interlayer to the interior of the infrared reflective layer **108** may preferably be darkened in color so that the infrared light reflected is not absorbed by the darkened layer. The glazing **100** may include a low-emissivity (low-e) coating **109** on a second glass sheet **102** for further improving the thermal comfort of a vehicle interior. When the

switchable layer **107** is in a relatively transparent state, the glazing **100** may have a total visible light transmission of 5% to 15% and total light transmission less than 25%.
(30) A glazing **100** without a switchable layer **107** may also include an infrared reflective coating **108** or a low-e coating **109**. Where a glazing includes a low-e coating **109**, it may be preferable that the second light extraction means is provided on a surface of the second glass sheet **102** opposite the low-e coating **109** (i.e., the third glass surface **113**). Where the second glass sheet **102** having the low-e coating **109** is also the second light guide **132**, the low-e coating **109** may have a neutral color reflection of light at the reflection angle within the second light guide **132**. If the color reflection is not neutral, a shift in the light color may occur.
(31) FIG. **8** illustrates a cross section of another example embodiment of a laminated glazing of the present disclosure. In some embodiments, the glazing **100** may include a first glass sheet **101**, with a first surface **111** and a second surface **1112**, and not a second glass sheet. For example, first light guide **122** may be configured to direct light to a second light guide **132**, which may be implemented as a glass sheet **101**, in order to supply ambient lighting to a passenger window of a vehicle.
(32) To provide a suitable surface for installation in a vehicle and to protect the lighting mechanism, the glazing may be encapsulated. The encapsulation material may be positioned around the glazing edge, over the lighting system, including the first light guide **122** and the first light extraction zone **123**.
(33) The above description of the disclosure is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the common principles defined herein may be applied to other variations without departing from the spirit or scope of the disclosure. Further, the above description in connection with the drawings describes examples and does not represent the only examples that may be implemented or that are within the scope of the claims.
(34) Furthermore, although elements of the described aspects and/or embodiments may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Additionally, all or a portion of any aspect and/or embodiment may be utilized with all or a portion of any other aspect and/or embodiment, unless stated otherwise. Thus, the disclosure is not to be limited to the examples and designs described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A glazing, comprising: a laminated structure including: a first glass sheet; a second glass sheet; and an interlayer between the first glass sheet and the second glass sheet; and a lighting system including: three first light guides; three first light sources, each corresponding to one of the three first light guides and attached to each corresponding first light guide; three first light extraction zones, wherein each first light extraction zone includes a first light extraction means for directing light into a second light guide within the laminated structure; and a second light extraction zone including a second light extraction means wherein light in the second light guide is extracted at the second light extraction zone.
2. The glazing according to claim 1, wherein the glazing is a sunroof.
3. The glazing according to claim 1, wherein the second light extraction means is on a surface of the first glass sheet or the second glass sheet.
4. The glazing according to claim 1, wherein the second light extraction means is on a film.
5. The glazing according to claim 4, wherein the film is laminated between a first glass sheet and a second glass sheet.
6. The glazing according to claim 4, wherein the film is applied to a surface of the first glass sheet or the second glass sheet.

7. The glazing according to claim 1, wherein the second light extraction means includes a print.
8. The glazing according to claim 7, wherein the second light extraction means is printed on a surface of the first glass sheet or the second glass sheet.
9. The glazing according to claim 7, wherein the second light extraction means is printed on a surface of the interlayer.
10. The glazing according to claim 1, wherein the second light guide is the second glass sheet and the second glass sheet faces a vehicle interior when installed.
11. The glazing according to claim 1, wherein the first light extraction means includes a luminescent composition.
12. The glazing according to claim 1, wherein the at least one first light source is a laser.
13. The glazing according to claim 1, wherein the at least one first light guide at least partially or completely surrounds the laminated glazing.
14. The glazing according to claim 13, wherein the at least one first light guide and the at least one first light extraction zone are adhered to an edge of the laminated structure and an encapsulation material is applied around an edge of the laminated structure, over the at least one first light guide and the at least one first light extraction zone.
15. The glazing according to claim 1, wherein the at least one first light guide is at least partially laminated between the first glass sheet and the second glass sheet and wherein the at least one first light extraction zone is laminated between the first glass sheet and the second glass sheet.
16. The glazing according to claim 1, wherein a light absorption coefficient in the second light guide is less than 0.1/cm.
17. The glazing according to claim 1, wherein the at least one first light extraction zone has at least one area of a variable light intensity.
18. The glazing according to claim 1, wherein light from the three first light extraction zones are three separate colors: red, green, and blue.
19. The glazing according to claim 1, wherein multiple first light extraction zones each include a different light extraction means.
20. The glazing according to claim 1, wherein the laminated structure further comprises a switchable layer.
21. The glazing according to claim 1, further comprising an infrared reflective layer.
22. The glazing according to claim 21, wherein the infrared reflective layer is a coating provided on the first glass sheet.
23. The glazing according to claim 21, wherein the laminated structure further comprises a switchable layer, and wherein the infrared reflective layer is provided as a film laminated between the first glass sheet and the switchable layer.
24. The glazing according to claim 1, further comprising a low-emissivity coating on the second glass sheet.
25. The glazing according to claim 1, wherein the first interlayer is darkened in color.
26. A glazing having lighting capabilities, comprising: a glass sheet; and three first light guides at least surrounding the glass sheet; three first light sources, each corresponding to one of the three first light guides and attached to each corresponding first light guide; three first light extraction zones, wherein each first light extraction zone includes a first light extraction means for directing light into a second light guide within the laminated structure; and a second light extraction zone including a second light extraction means wherein light in the second light guide is extracted at the second light extraction zone.
27. The glazing according to claim 26, wherein the at least one first light guide completely surrounds the glass sheet.
28. The glazing according to claim 26, wherein the glazing is a sunroof.
29. The glazing according to claim 26, wherein the second light extraction means is on a surface of the glass sheet.

30. The glazing according to claim 26, wherein the second light extraction means is on a film applied to a surface of the glass sheet.
31. The glazing according to claim 26, wherein the second light extraction means includes a print.
32. The glazing according to claim 26, wherein the first light extraction means includes a luminescent composition.
33. The glazing according to claim 26, wherein the at least one first light source is a laser.
34. The glazing according to claim 26, wherein the at least one first light guide and the at least one first light extraction zone are adhered to an edge of the glass sheet and an encapsulation material is applied around a glass sheet edge, over the at least one first light guide and the at least one first light extraction zone.
35. The glazing according to claim 26, wherein a light absorption coefficient in the second light guide is less than 0.1/cm.
36. The glazing according to claim 26, wherein the at least one first light extraction zone has at least one area of a variable light intensity.
37. The glazing according to claim 26, wherein light from the three first light extraction zones are three separate colors: red, green, and blue.
38. The glazing according to claim 26, wherein multiple first light extraction zones each include a different light extraction means.
39. The glazing according to claim 26, further comprising a low-emissivity coating on the second glass sheet.